

cramped little shop that probably doesn't even have a name board outside. But shops like these stock most of the hardware and the prices too are often lower than those in the large retail shops. The final, and probably the most popular place to buy your hardware, is the grey market. Here you will find every component you need at prices that are at least 20 per cent lower than what is offered in retail outlets. But buy at your own risk, for the components will not have any warranty and the product might not come with any documentation. Choose this option only if you have enough knowledge of computer hardware or else you could end up with spurious or substandard components.

When is the right time to buy? That depends on how long you can wait. On the processor front, AMD is a great buy because it offers a performance that is very good at its price. However, Intel is all set to change that. You can now purchase a 1.8 GHz P4 for Rs 12,000, whereas a 1.4 GHz AMD Athlon processor (which is similar in performance to the 1.8 GHz P4) is available for Rs 9,500. After the change in prices the Intel processor seems a better buy because it can be over-clocked far more than the AMD without getting overheated.

You also have plenty of choices in motherboards—you can choose from among at least five motherboard brands for every processor. As for hard disks, 10 GB is nothing these days; you can now have an entry-level hard drive with a capacity of 20 GB priced as low as Rs 4,500. RAM prices have fallen to an all time low of Rs 1,500 for 256 MB and everyone is cashing in on this by packing their system to the gills with truckloads of RAM.

Elbow grease

After you've got all your hardware together, pick up a screwdriver set and get ready to set up your brand new system.

Connecting the motherboard: Get rid of any static charges by touching the plain body of the cabinet. Remove the back panel from the cabinet, and mount the motherboard onto it. See that it is correctly aligned with the panel. To avoid short



Remove the back panel from the cabinet and connect the motherboard to the panel

How about an Assembled PC?

Some of you who are a little hesitant about building your own machine, and not wanting to buy a branded PC either, often go in for an assembled one. There are several self-proclaimed assemblers who will build a PC on the basis of the configuration you give them. You can get your PC in a week's time and you will have to pay some extra charge for assembling. Seems good, doesn't it? But consider this. If you don't have any knowledge on the hardware front, there are chances that the assembler might take you for a

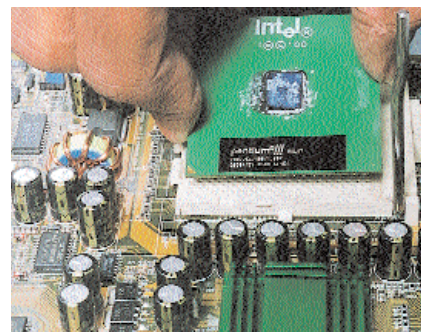
ride by giving you, for instance, a skimpy 4 MB SiS card, which will not let you play high-end 3D games. Often the one-year warranty on servicing and replacement of components is not honoured promptly enough. You may have to make a number of phone calls for weeks on end to get any sort of remedy to your problems. More importantly, the software that these assemblers provide are usually pirated—no licensed ones here. So you could get caught for using pirated software while the assembler gives you the slip.

circuits, make sure that there is no direct contact between the motherboard and the panel. Take care to see that the washers have been properly attached between the motherboard and the metal panel.

Connecting the processor: Now, connect the processor to the motherboard. Processors currently available come with a socket 370 interface (for the Pentium III and the Celeron), a socket 423/478 interface (for the Pentium 4), or a socket A interface (for the Athlon/Duron).

The socket interface is one of the most common interfaces available today. It is somewhat rectangular in shape and has a lever on its side, which must be lifted to open it. Every processor has a small indication as to the manner in which it is to be placed on the motherboard. On most processors, you will find a marking on the corner in a particular shape and a complementary mark on the motherboard socket. Insert the processor according to the mark and push the lever down. Do not force the processor into the socket as this might damage the pins and render the processor useless. You will sense some resistance on the lever while closing it, but this is normal and once closed, it will hold the processor in place.

Attaching heat sink and fan (for socket interface processors): You will have to attach the heat sink and fan to the processor to keep the processor's temperature under control. You need to apply some thermal paste on the processor's surface and some on the heat sink too. Once this is done, gently place the heat sink over the processor taking care that there are no air bubbles between the processor and the heat sink because if the contact is poor, it might heat the



Mounting the processor the motherboard



Attach the heat sink to the processor